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Bridging the Silence

A Culturally Grounded Digital Platform for End-of-Life Communication and Palliative Care Navigation in China

Jason Shi · Ksenia Kozhukhar · Antonio Capilla
Selena Zhang · Carlina Moore

TA: Michelle Shi

Research Center for Culture and Psychology, Tsinghua University

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Abstract. When an older Chinese adult is found dead alone in their flat days after the fact, the response that follows is almost always about detection — guardianship paperwork, community check-in rounds, an app that asks users each day whether they are still alive. Almost none of it is about company. Detection is the tractable end of the problem because the aloneness, not the death, is what reads as shameful in a culture organised around filial duty, and shame is cheaper to repair in the account than in the situation. This proposal takes that asymmetry as its central image, because Chinese families facing a terminal illness perform a smaller version of it daily: a prognosis withheld from the patient to protect them, while the patient, sensing it, protects everyone else by not asking. Three silences stack — a society without public vocabulary for dying, a family holding a secret everyone already knows, and a service system that has grown faster than public awareness of it, with national pilot programmes now covering 185 cities and districts while, in a self-selected online sample that skewed female and university-educated, only two in five respondents could explain what hospice and palliative care is — a figure that is more plausibly a ceiling than an average. We propose 渡 (Bridge), a bilingual WeChat platform combining guidance on how to talk about death, drafted for clinical review and not yet signed off, a navigable directory of palliative services, a moderated peer and specialist community, and a deliberately limited AI assistant whose only success condition is a completed handoff to a person. We set out the design and its theoretical derivation, a formative interview study with older Chinese adults that will revise it, a model of mechanisms, and a phased evaluation agenda with pre-specified conditions under which we would conclude we are wrong. A working prototype exists; nothing here has been tested.

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Keywords. End-of-life communication; palliative care; hospice; China; protective concealment; death education; digital health; family caregivers; human–AI interaction.

Declarations. *Ethics approval:* none sought. No data have been collected from human participants; the study described in §6 and §8 is proposed, not conducted, and would require institutional and site approval before recruitment (§8.4). *Funding:* none. *Conflicts of interest:* the authors designed the platform evaluated in this proposal and are therefore not independent evaluators of it; the mitigations we commit to are stated in §8.4. *Data availability:* no data were generated; the prototype is a self-contained file available from the authors on request.

1 Introduction and Background

1.1 Two rooms

NANJING, JUNE 2019 A woman in her seventies who lived alone was found dead in her flat. She had a son and a daughter who came every couple of weeks; the person who found her was a community worker making a routine visit, and she was lying beside her bathtub. A month later, in Shanghai's Baoshan district, a 67-year-old man who lived alone and had left his front door unlocked was not discovered until the end of July, when the summer heat and the smell prompted his neighbours to call the police. Both cases were reported in the local press, and then in the regional English-language press (South China Morning Post, 2019).

Neither case was unusual enough to be a scandal, which is the point. Nearly sixty per cent of Chinese people over sixty now live in what are called empty-nest households, alone or with only a spouse — a share that rose from under half in 2010 across the decade to 2020, on the fifth national survey of the living conditions of urban and rural older adults (Yicai Global, 2024). Solitary deaths discovered days or weeks afterwards recur often enough to have generated their own industry of responses: notarised guardianship agreements, community check-in rounds, and, in 2026, a mobile application whose entire function is to ask its users each day whether they are still alive (The Diplomat, 2026).

What is striking about that response is what it is for. Every part of it addresses *detection*; none of it addresses company. And detection is the tractable end of the problem because the aloneness is the shameful part. Death itself is public, ritualised, and extensively provisioned with custom — ancestor worship, grave-sweeping, mourning obligation (Hsu et al., 2009) — whereas under filial obligation (孝) a parent's solitary death reads as the child's public failure. When shame attaches to the circumstances rather than to the fact, the cheapest available repair is not to the circumstances. It is to the account of them.

That is the move this proposal wants to name and then interrupt: *repairing the story instead of the room.* It costs nothing, it protects everyone's face, it is performed collaboratively and kindly, and it leaves the situation exactly as it was. The version of it that concerns us is smaller and more intimate than a news report — it happens at a bedside, weeks before anybody dies — and we return to it at each stage of the argument, because every part of this problem turns out to have one.

Because the phrase carries a good deal of weight in what follows, we fix its meaning once. *Repairing the story instead of the room* denotes a class of response in which shame attaches to the circumstances of a situation rather than to the facts of it, so that the cheapest available repair is directed at the account rather than at the situation — and the situation is therefore left intact. It is not a metaphor we extend as convenient; it is a mechanism with a condition of application, and where the condition does not hold we say so.

Two qualifications, stated here rather than deferred to §10. The two cases are journalism, not data: they illustrate a pattern without establishing its prevalence. And they are not, strictly, instances of our subject. Those people died alone because nobody was in the room; the families this proposal concerns are in the room and silent. What the cases share with protective concealment is the mechanism, not the situation, and a reader who finds the opening too convenient may discount it without discounting §2. The same caution applies to the empty-nest and smartphone-use figures we draw from press reporting of the national survey of older adults' living conditions: they are indicative, and the design decisions that rest on them (§4.5) should be read as provisional until the underlying survey is consulted directly.

1.2 Death, dying, and what may be said

Chinese attitudes toward death are not simply avoidant. They are the sediment of several traditions at once — ancestor worship, Confucian ethics, Daoist naturalism, Buddhist cosmology, traditional medicine — which together treat death as a natural passage while making the family, not the individual, the unit that owns it (Hsu et al., 2009). Talk about death in the abstract is not forbidden; a great deal of custom exists precisely to structure it.

What is constrained is something narrower and more consequential: talk about *this* death, with *this* person, *now*. Clinicians describe the taboo as a live obstacle to conversations about resuscitation and goals of care (Cheng et al., 2019); nurses describe managing a conflict between the professional norm of honesty and the cultural norm of protection (Tu et al., 2022); and medical educators report that the same attitudes shape what can be taught (Willemsen et al., 2021).

The evidence also contains a useful corrective. In focus groups with older Chinese immigrants in Canada, most participants discussed death readily, using either the plain word or a euphemism, and raised treatment, end-of-life care, and preparation without prompting (Zhang, 2022). If older people are more willing to talk than their families assume, then the taboo may be less a property of individuals than a property of situations — a collective expectation that each person maintains on everyone else's behalf, exactly as the neighbours and children in those reports maintained arrangements that were kind, ordinary, and insufficient. That possibility is central to our design, and it is the first thing our formative study (§6) is built to test rather than assume.

1.3 Demographic and service context

China is ageing quickly. At the 2020 census 190.6 million people, 13.50% of the population, were aged 65 or over (National Bureau of Statistics of China, 2021); by the end of 2024 the figure was 220.23 million, or 15.6% (National Bureau of Statistics of China, 2025). The burden of cancer and other life-limiting illness is rising with it. Most dying happens at home, and most of the care is given by relatives with no training and no relief.

Provision is expanding. Three rounds of national hospice and palliative care pilot programmes, beginning with five cities in 2017 and extended in 2019 and 2023, now cover 185 cities and districts (Lu et al., 2024). But in a nationwide survey of the general public, only 41.1% of respondents could explain what hospice and palliative care is (Tang et al., 2025). That figure deserves its caveat rather than a footnote: the sample was recruited online and self-selected, 77.8% were women and 52.5% held an undergraduate degree, and the authors themselves note that voluntary participation is likely to have drawn respondents with more sympathy for the topic. It is therefore better read as a ceiling than as a population mean, which strengthens rather than weakens the argument that follows. Low utilisation alongside expanding provision is reported in the same paper by way of earlier service-side work rather than from its own data (Wang et al., 2018; Chung et al., 2021). Reviews of the field in mainland China and the wider region find the same pattern from the service side: research and provision concentrated in a few urban centres, palliative care poorly integrated into primary care, and informal caregivers consistently reporting unmet needs for education and for reduction of burden (Wang et al., 2018; Chung et al., 2021). Even within intensive care, clinicians report gaps in palliative knowledge and practice (He et al., 2025), and patients' own attitudes turn out to depend heavily on what they have been told rather than on fixed belief (Zhang et al., 2025).

So the gap is not only a shortage of services. Increasingly, help exists and cannot be found, named, or asked for — and the people who would benefit are maintaining a silence that prevents them from asking.

1.4 Aim and contributions

We propose 渡 (Bridge): a bilingual, culturally grounded platform for end-of-life communication and palliative navigation in China, delivered as a WeChat Mini Program. Our contributions are five.

1. A problem formulation — three stacked silences and the repair-the-story mechanism that sustains them (§2).
2. The platform concept and its theory-to-feature derivation, including an explicit third position for AI in mental health: not clinician, not companion, but bridge (§4).
3. An honest scope boundary stating what the platform does, what is deferred, and what cannot be solved by software at all (§4.7).
4. A formative interview study with older Chinese adults and their families, with design consequences committed to in advance (§6).
5. A model of mechanisms and a phased evaluation agenda with pre-specified disconfirming conditions (§7–8).

A working prototype exists and is described throughout. It has not been clinically reviewed, no study has been run, and every effect in this document is hypothesised.

2 The Problem

2.1 Silence one: dying is absent from public discourse

Death is ritually present in Chinese life and practically absent from it. There is elaborate custom for after a death and very little language for the weeks before one. Death education is thin: in a survey of community-dwelling Chinese older adults, attitudes toward death were associated with a substantial expressed *demand* for death education, which is to say that the people closest to the topic are asking to be taught something nobody is teaching them (Lei et al., 2022).

The consequence is a vocabulary problem. A person just told that their mother has months to live does not know that what they want is called 安宁疗护, that it may appear locally as 舒缓治疗 or 临终关怀, or that it is something one may request. They cannot search for what they cannot name.

The vocabulary has since been standardised, which sharpens rather than dissolves the problem. The 2023 edition of 《常用临床医学名词》, promulgated by the National Health Commission (National Health Commission of the People's Republic of China, 2024), fixes 缓和医疗 as palliative care, delivered from diagnosis onward, and 安宁疗护 as hospice care at the end of life; the Commission has used 安宁疗护 as its term of art since the first pilot round in 2017 (Lu et al., 2024). A national expert consensus published in 2025 built definitions for both terms on that standard, noting that inconsistent terminology had until then hindered policy, practice and research alike (Chinese Expert Consensus Working Group on the Definition of Palliative Care, 2025). Standardisation binds institutions, not families. A person searching in the week after a diagnosis still meets whichever term their hospital happens to use, and now meets two official terms where there was one.

2.2 Silence two: protective concealment inside the family

Disclosure in Chinese oncology has historically been family-mediated rather than patient-directed: relatives are told first and decide what the patient will hear, and patients and families hold measurably different views about how much should be said at which stage of illness (Jiang et al., 2007; Cheng et al., 2019). The intention is protective. The effect is a household in which everyone knows, nobody speaks, and each person believes the silence is their own gift to the others.

This configuration has a name. Glaser and Strauss (1965) distinguished four awareness contexts around a dying person — closed awareness, suspected awareness, mutual pretence and open awareness

— and described mutual pretence as a collaborative performance sustained by rules that neither party acknowledges. The Chinese idiom 心照不宣 names something adjacent but not identical: a tacit shared understanding, without the implication of a sustained performance. We use the technical term for the configuration and the idiom for how families themselves describe it.

Communication Privacy Management theory explains why this is so stable (Petronio, 2002). Private information is held inside collectively managed boundaries with rules about who may know and who may tell. Concealment is not a lie an individual tells; it is a boundary rule a family maintains. Breaking it is not a change of mind but a breach of an agreement — which is why exhortation to "just be honest" fails, and why the useful intervention is a script for renegotiating a rule rather than an argument for abandoning it.

The theory also predicts what happens when renegotiation goes badly. Petronio calls it boundary turbulence: an attempt to reset a rule can produce retaliation, tighter boundaries, and exclusion of whoever moved first. This is not an incidental risk of our intervention; it is what the theory says a proportion of attempts will cost. It is also why the adverse events we pre-specify in §8.4 — family conflict after an attempted conversation, disclosure the patient did not want — are theory-derived predictions rather than a defensive list. A platform that provokes some turbulence is behaving as the model says it should; one that provokes none has probably not been used.

The mechanism Nobody in Nanjing or Baoshan conspired. The children's fortnightly visits were, by their own lights, filial; the community worker's round was the safety net working as designed. Protective concealment operates the same way at the scale of one bedroom: several people, each acting generously, produce a silence that none of them would have chosen alone and that none of them can now be the first to break.

2.3 Silence three: the system is invisible

Where services exist, families frequently cannot find them, and when they do they cannot tell whether the listing is current, whether home visits are available, or whether anything is covered. Reviews of mainland provision describe uneven distribution and weak integration with primary care (Wang et al., 2018; Chung et al., 2021), against low awareness and uptake (Tang et al., 2025). A caregiver has a limited number of afternoons of energy, and a wrong phone number costs one of them.

2.4 Who bears the cost

The three silences distribute their costs unevenly (Figure 1). The *patient* is isolated, may have symptoms under-treated because the conversation that would surface them never happens, is excluded from decisions about their own body, and does not get to say goodbye. The *caregiver* carries the whole picture alone, decides without support, and enters bereavement having been the only informed person in the family — a configuration we would expect to raise the risk of complicated grief. We should be careful about that expectation: Prigerson et al. (1995) supplies the instrument by which complicated grief is measured, not evidence that this particular configuration produces it, and we know of no study that has tested the link directly. The conjecture stands on its own, and Phase 4 (§8.2) is where it would be examined, and one consistent with the unmet needs for education and burden reduction reported across the region (Chung et al., 2021). The *child* is told nothing in order to be spared and fills the gap from imagination, frequently with the conclusion that it is their fault. None of the three is addressed by information alone.

2.5 What is missing

Two gaps follow, and stating them requires first stating what exists. China has a Chinese-language digital and voluntary presence in this field, and it is older than the platform we propose. 选择与尊严 (Choice and Dignity) has published advance-directive material online since 2006, offering a Chinese adaptation of the *Five Wishes* document, and gave rise to the Beijing Living Will Promotion Association

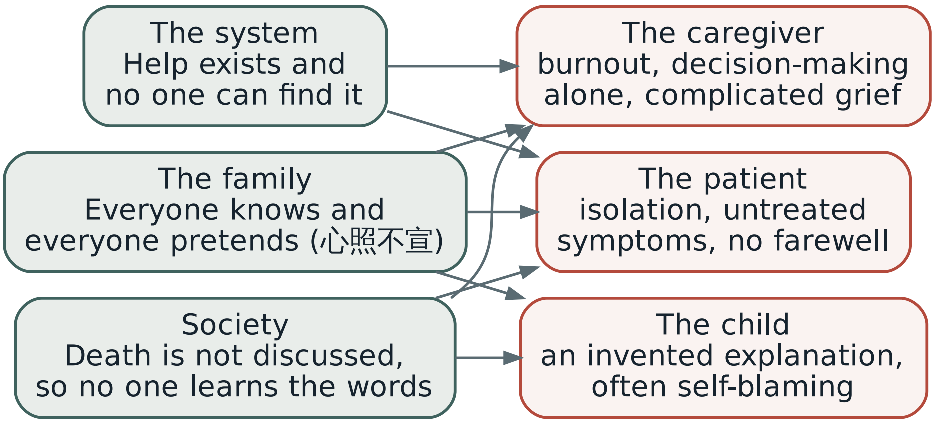

Figure 1: The three silences and how their costs fall on the three parties. Each silence is addressed by a different component of the platform (§4.3).

in June 2013 (Beijing Living Will Promotion Association). 手牵手生命关爱发展中心 has trained volunteers and supported bereaved families in Shanghai since 2008 (Hand in Hand Life Care Development Centre). And the Li Ka Shing Foundation opened the mainland's first 宁养院 at Shantou University Medical College in 1998, scaling it into a national home-based hospice programme from January 2001 (Li Ka Shing Foundation). What we have not found is a platform that combines communication guidance, service navigation and a moderated community in a single surface, aimed at the family member who currently holds the prognosis alone. Existing international digital work in palliative care also recruits largely through hospices, which builds in two assumptions that fail here: that the family is already inside a palliative service, and that the diagnosis is shared knowledge within the family. Second, work on AI in mental health has concentrated on assessment and therapy-like interaction and has little to say about end-of-life contexts. We return to this in §3.5.

How we searched. Both claims in this subsection are negative, and a negative claim is only as good as the search behind it, so we state ours. Between June and August 2026 we searched PubMed, Scopus, PsycINFO, the ACM Digital Library and CNKI, in English and in Chinese, combining terms for the setting (*palliative*, *hospice*, *end-of-life*, *bereavement*, 安宁疗护, 缓和医疗) with terms for the artefact (*app*, *mHealth*, *digital intervention*, *WeChat*, *mini program*) and, for the second claim, with terms for the agent (*chatbot*, *conversational agent*, *large language model*, *companion*). We restricted to work published from 2010 onward, screened titles and abstracts, and followed citations forward and backward from the sources retained. We also searched the Chinese-language web and the WeChat Mini Program directory for deployed services, since a good deal of practice in this field never reaches a journal. The search was not systematic in the PRISMA sense and we do not present it as one; it is a structured scan, and the claims it supports are correspondingly phrased as *we have not found* rather than *there is not.*

Table 1 sets out what the search did return among the Chinese-language actors, so that a reader can judge the gap rather than take our word for it.

Two assumptions built into the international work are worth isolating, because they are what make it inapplicable here rather than merely foreign: that the family is already inside a palliative service, and that the diagnosis is shared knowledge within the family. Both fail at exactly the moment our primary user arrives.

Table 1: Chinese-language provision in this field, and the functions each covers. The gap we claim is the combination in a single surface aimed at the family member holding the prognosis, not the novelty of any one function.

Actor	What it does	Words guidance	Map navigation	Peers community
选择与尊严 / Beijing Living Will Promotion Association	Advance-directive material and a Chinese *Five Wishes*; advocacy and living-will registration	partial	—	—
手牵手, Shanghai	Volunteer training, hospital companionship, bereavement groups; offline and relationship-based	partial	—	offline
Li Ka Shing Foundation 宁养院 network	Free home-based hospice care at partner hospitals; a service, with a directory of its own units	—	partial	—
Hospital and municipal directories	Lists of accredited 安宁疗护 institutions, published administratively and rarely dated	—	partial	—
International palliative-care apps	Symptom diaries, ACP tools, caregiver education; recruit through hospices	yes	partial	rare
渡 (proposed)	All three above one thin data layer, entered without naming the need, and aimed at the family member who holds the prognosis	yes	yes	yes

3 Theoretical Foundations

3.1 Death attitudes and avoidance

Terror Management Theory holds that mortality awareness generates anxiety managed through cultural worldviews and avoidance (Greenberg et al., 1986). On this account the taboo is not an absence of thought about death but a mechanism for not having to have it — functional for the individual, costly for the family. The theory also distinguishes proximal defences, which are conscious and suppress the thought, from distal symbolic defences, which are not (Pyszczynski et al., 1999); a guide read deliberately at two in the morning engages the former, and that is the level at which we claim to act.

Death attitudes are multidimensional rather than simply high or low: fear of death, death avoidance, and several forms of acceptance are separable (Wong et al., 1994). Templer (1970) is the earlier unidimensional instrument that the Death Attitude Profile–Revised was built to supersede, and we cite it for that lineage rather than for the multidimensional claim. The distinction matters for measurement (§8.3), because an intervention that reduces *avoidance* without reducing *fear* would still be doing its job.

The prediction that runs against us. Terror Management Theory's most replicated finding is not avoidance but worldview defence: mortality salience strengthens adherence to the cultural worldview a person already holds (Burke et al., 2010). In this setting the worldview being defended is filial obligation and the protective concealment it licenses. Read strictly, then, the theory predicts that a platform which raises mortality salience — as ours does on almost every screen — may *harden* the norm it is built to loosen.

We take that seriously rather than setting it aside, and our answer is the subject of §3.2: we do not ask anyone to abandon 孝, we supply it with a second available expression at the end of life, so that speaking first can be performed as care rather than as surrender. The threat-lowering apparatus of the Lantern Language System (§4.2) is what is supposed to make that reframing reachable before defensive processing begins. But this is an argument, not a result, and we therefore add it to the disconfirming conditions in §7: if death avoidance *rises* rather than falls in the subgroup with the strongest filial orientation, worldview defence has beaten our framing, and the mechanism we proposed is wrong in a way that more content would not fix.

3.2 Filial piety, and why speaking first feels like giving up

Filial obligation supplies the moral weight that holds the second silence in place. The duty to protect and sustain a parent makes the adult child the natural decision-maker, and simultaneously makes raising the prognosis feel like being the person who abandoned hope on the parent's behalf. Family structure and the identity of the family decision-maker are empirically associated with end-of-life preferences (Tang et al., 2025), and the family — not the patient — is the culturally recognised owner of the situation (Hsu et al., 2009). Filial piety therefore plays a double role: it produces the care, and it produces the silence. An intervention that treats it only as an obstacle will be rejected by the people it is for.

3.3 What is known to help

Children understand death through separable components. The classical review identifies three — irreversibility, non-functionality and universality — with most healthy children in urban-industrial settings achieving all three between five and seven years of age (Speece and Brent, 1984); causality was added as a fourth in later work (Speece and Brent, 1992). The developmental sequence is not culturally invariant, and the comparison that matters here has been made directly: Brent et al. (1996) traced the concept across Chinese and United States children aged three to seventeen. Guidance that ignores this either overwhelms a child or leaves them to construct their own account — and in the younger bands that account is often magical, the child concluding that their own thoughts or wishes caused the death. In a qualitative study of 157 children bereaved of a parent, Christ (2000) documents that pattern and identifies reassuring the child that they are not to blame as one of the surviving parent's central tasks, which is the developmental case for honest, age-calibrated involvement rather than protective exclusion.

The dying benefit from interventions targeting meaning and being known rather than information transfer: dignity therapy, life review, and legacy work are established brief approaches with this shape (Chochinov et al., 2005). These approaches share a shape — attention to the relationship rather than to the disease — and one of them has been compressed into a form short enough to hand to a family. It is the form our content is built on, and it arrives in China already localised.

Caregivers are best understood through a stress-process model, in which objective demands are mediated by appraisal and available resources rather than determined by workload alone (Pearlin et al., 1990). This is what makes a communication and navigation intervention plausible at all: we cannot change the illness, but appraisal and resources are reachable.

The tested intervention our checklist most resembles. The First Ten Days checklist is phrased throughout as things to *ask* rather than things to know, and that is not a novel idea: structurally it is a Question Prompt List. QPLs have randomised evidence in oncology and palliative care — a prompt list given to patients with advanced cancer and their caregivers increased question-asking about prognosis and end-of-life care without lengthening the consultation or raising anxiety (Clayton et al., 2007), and the intervention class has been reviewed systematically (Brandes et al., 2015). Structured conversation guides for clinicians have comparable support (Bernacki et al., 2019).

Our contribution is therefore not the format. It is that every tested version of this format sits inside a clinical encounter, is handed over by a clinician, and assumes the diagnosis is already shared. We propose to move the same mechanism upstream of the consultation and into the family, in a culture where the second of those assumptions routinely fails — a smaller and more defensible claim than novelty, and a testable one.

There is also a cautionary literature we should cite against ourselves. Advance care planning interventions have a long record of improving documentation and communication self-efficacy while failing to move patient-level outcomes, to the point that the field has been asked publicly whether the intermediate gains mean anything (Morrison et al., 2021). That is precisely the failure mode §7 names, and a further reason our confirmatory endpoint is chosen carefully rather than aspirationally (§8.2). For **the bereaved**, the Dual Process Model describes healthy grieving as oscillation between loss-oriented and restoration-oriented coping (Stroebe and Schut, 1999) — implying that a platform should not push continuous engagement with grief, a point we take literally in §4.4. One extrapolation should be flagged rather than smuggled: the model was formulated for the bereaved, and our primary user is a caregiver in anticipatory grief, before a death. Later work extends the account to the family system and to coping that unfolds around a dying person (Stroebe and Schut, 2015), which is the ground we stand on, but the transfer is an assumption and belongs with the others in §7.

3.4 Relationship completion, and the version China already has

Among the brief interventions that target meaning rather than information transfer, one has a distinctive property: it reduces to four sentences. Sinophone hospice practice knows it as 四道人生 — 道谢, 道歉, 道爱, 道别: to thank, to apologise, to express love, to say farewell. Clinical sources credit its formulation to 趙可式, the nurse-scholar regarded as the founder of hospice palliative care in Taiwan and instrumental in the passage of the 安寧緩和醫療條例, and it travelled from Taiwan into Hong Kong and mainland practice (Hospice Foundation of Taiwan, a; Comprehensive Handbook for Cancer Patient Caregivers). It is taught alongside the 四全 model of whole person, whole family, whole course and whole team — to which a fifth, whole community (全社區), was later added (Hospice Foundation of Taiwan, b). We should note that the documentation for both frameworks is clinical and educational rather than a research literature, which is a limitation of the evidence base and not only of our reading of it. Its international counterpart is Byock's account of relationship completion, whose four things are thank you, please forgive me, I forgive you, and I love you (Byock, 2004).

The two lists differ, and the difference has a history worth recording. 四道 pairs an apology with a farewell, where Byock pairs asking forgiveness with granting it; the Anglophone list has no explicit goodbye and the Sinophone list, as originally formulated, had no explicit absolution. Hong Kong palliative practice subsequently added a fifth — 道諒, to forgive — on the reasoning that 道歉 covers only the asking, and 五道 is now the ordinary usage there (Comprehensive Handbook for Cancer Patient Caregivers). So the tradition did not simply diverge from Byock; it noticed the same gap and closed it from the other side, arriving at five where the Anglophone framework has four. That convergence is stronger evidence for our purposes than a clean contrast would have been: it suggests the components are recognisably universal while their ordering and emphasis are not, which is precisely the claim §8.1 (RQ3) proposes to test.

The mechanisms are the ones already invoked above. The statements enact in miniature what life review and dignity-oriented work do at length: they make the relationship, rather than the illness, the subject of the remaining time (Chochinov et al., 2005). Their absence is what the bereavement literature calls unfinished business, and their presence supplies the material from which a continuing bond with the deceased is later constructed (Klass et al., 1996) and from which the survivor reconstructs a meaning for the loss (Neimeyer, 2001). This is why the framework is not only for the dying. It is scaffolding for the person who will be left.

Its value in a Chinese setting follows from the account in §2.2. Under protective concealment, a family may not say *you are dying*; the prohibition attaches to the prognosis, not to the relationship. Gratitude, apology, love and farewell can all be spoken without naming what everyone already knows, and in a relational ethics they read as the repair of harmony rather than as individual emotional disclosure. The framework therefore enters the hardest conversation through the side door. That is also its limit: it can complete a relationship inside a silence it does not break, and whether that is sufficient is an empirical question we do not answer here.

In the platform, the framework structures the script cards in the knowledge hub (§4.3). It does not structure the age-graded child guides, which derive from the components of the death concept, and it does not structure Rehearse, which remains unbuilt (Table 3). We present 四道 as four and note

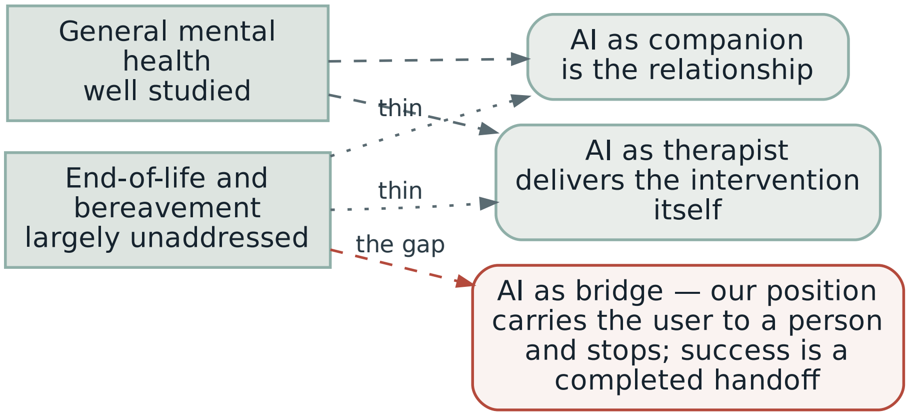

Figure 2: Positioning relative to existing paradigms for AI in mental health. The unaddressed region is both the end-of-life context and the bridging role.

the Hong Kong fifth rather than adopting it, because the formative study (§6) should tell us which the mainland families we serve recognise.

3.5 AI in mental health, and the position we take

Two roles for language models are now technically available. The first is AI as clinician or assessor: multi-agent systems conduct structured psychiatric-style interviews and reason over diagnostic criteria (Bi et al., 2025). The second is AI as companion or therapeutic presence, including avatar-based approaches in family therapy (Colvin and Benjamin, 2025). Both carry risks that are now well rehearsed in the literature and that we take as given: fluent fabrication of clinical content, dependency and displacement of human relationships, and unreliable behaviour in precisely the crises where reliability matters.

The gap we identify is positional (Figure 2). Both established roles place the machine *inside* the helping relationship. Work positioning the machine as a scaffold rather than as a party to the relationship does exist in adjacent settings — notably systems that support human peer supporters rather than replacing them (Sharma et al., 2023). What we have not found is that stance applied to end-of-life or bereavement contexts, or stated as a success condition rather than as an architecture. That third position is the design commitment of §4.1, and it is falsifiable: it predicts that a system optimised for handoffs outperforms one optimised for engagement.

A claim advertised as falsifiable should say what would falsify it, and where. The comparison is not free: an engagement-optimised comparator means deliberately deploying retention mechanics to people about to be bereaved, which we are unwilling to do at any scale. What is affordable is the vignette study. Phase 2a (§8.2) therefore carries a second factor in which identical content is framed either for handoff — ending in a named human route — or for engagement — ending in an invitation to return, with progress and continuity cues. The outcome is stated intention to contact a human service within the coming week, with perceived helpfulness secondary. If the engagement framing produces equal or greater intention to contact a person, the positional claim of this subsection is wrong and the restraint built into §4.4 is costing users something for nothing. That is the whole of the falsification, and we would rather state its limits than leave the word *falsifiable* doing unearned work.

4 The Proposed Solution: the 渡 Platform

4.1 Design philosophy: the bridge, not the destination

渡 means to ferry across. It names a crossing and, in Buddhist usage, a passage made with help. The platform is the ferryman: its value is entirely in the crossing, and none of it is in being a place to stay.

Buber's distinction between the I–Thou relation, in which another is encountered as a subject, and the I–It relation, in which the other is a means, gives this a precise form (Buber, 1937). A machine can only be an It, and a machine that successfully occupies the place of a Thou has displaced the relationship it was meant to serve. The failure mode is not an AI that performs badly. It is an AI that performs well enough that the difficult human conversation never happens — a new and very efficient way to repair the story instead of the room.

> *AI is a bridge, not a therapist. Every path the system offers ends at a person.*

The second commitment is that the family, not the individual, is the unit. Western end-of-life technology assumes an autonomous patient documenting preferences; we assume a family negotiating what may be said, and we design for the member who currently carries the information alone.

4.2 From theory to features

Table 2 states the derivation of each component. The intent is that no feature exists without a reason that could turn out to be wrong.

4.3 Architecture and components

Two constraints shape the build. The platform is designed as a WeChat Mini Program, though the present artifact is a self-contained web prototype and has not been through operator review: no installation, opens from a link inside the family group chat where care is already being coordinated, runs on cheap devices. The cost is a dependency on the operator's review rules and data handling. And the user arrives without vocabulary, which yields the governing rule: *the user must never have to name what they need in order to find it.* Search exists; it is never the primary path.

Four surfaces and two persistent utilities sit above a thin service and data layer, with every route designed to terminate in a person (Figure 3).

Home (渡口). Ten ordered steps on one screen, progress held locally, no account. The steps are phrased as things to *ask* rather than know: what the goal of treatment now is; who the single named contact is; what palliative provision exists locally; pain and symptom control as its own conversation; who else needs to know and who will tell them; the children; the paperwork; what help is available at home; one thing arranged for oneself; one person to be honest with. The sequence carries the burden of knowing what comes next, and at least three steps are simultaneously logistical and communicative — which is how a checklist smuggles a communication intervention into a tool that presents itself as practical. Four gates below it, written as sentences people actually say, route onward (Figure 4).

Words (言语). The knowledge hub, whose flagship track is not "what is palliative care" but *how to talk about death*, organised as two conversations. The child track pairs what a child understands at each age band with what the adult should therefore do, and prohibits two phrases explicitly: "went away" and "fell asleep." The adult track offers three scripted openings, names mutual pretence (心照不宣), lists what closes a conversation and what keeps it open, and presents 四道人生 (§3.4) with the note that there is no correct order and that starting is the whole of it.

Care Map (寻路). A filterable directory — hospice wards, palliative units, home-care teams, pain clinics, NGOs, offline groups — each listing carrying population served, home visits, insurance status, referral route, and a *last confirmed* date. Two hard problems here are not technical. Cold start forces

Table 2: Construct → design principle → feature.

Construct	Design principle	Feature
Death anxiety and avoidance (Greenberg et al., 1986)	Lower the threat of the topic before asking for behaviour	Anti-stigma framing; ordinary language; the child track as a low-stigma entry
Privacy boundaries as family rules (Petronio, 2002)	Supply a script for renegotiating a rule, not an instruction to break it	Conversation openings; the checklist step that assigns *who tells whom*
Filial obligation (Hsu et al., 2009; Tang et al., 2025)	Reframe speaking first as care rather than as surrender	Explicit section on mutual pretence and 孝
Components of the death concept (Speece and Brent, 1984)	Match language to what a child can hold at this age	Age-graded guides (3–6, 7–12, 13+); two prohibited phrases
Dignity and being heard (Chochinov et al., 2005)	Make the meaningful conversation concrete and finite	Script cards after 四道人生 (§3.4), inspired by dignity-oriented work rather than implementing it; saved cards
Caregiver stress process (Pearlin et al., 1990)	Act on appraisal and resources, since the illness cannot be changed	Ten-step checklist; care map; peer community
Self-efficacy (Bandura, 1997)	Give sentences, not principles; allow rehearsal	Verbatim openings; Rehearse mode (deferred, §4.7)
Oscillation in grief (Stroebe and Schut, 1999)	Never require continuous engagement	No streaks, notifications, or re-engagement mechanics

city-by-city coverage, stated honestly rather than implied. Decay is worse: a directory that is wrong in a crisis is more harmful than no directory. Our answer is a three-source maintenance loop — partner verification, user reports of what happened when they called, and a visible freshness indicator — which converts maintenance into a community task at the price of requiring moderation capacity from day one.

Forum (同行). Seven sections, pseudonymity by default, badges distinguishing anonymous members, role labels, and verified specialists. Its distinctive mechanisms are time-boxed specialist threads, in which a verified clinician explains how decisions get made while declining to advise on individual cases, and a professionals-only section for peer supervision. The forum is where the words are said aloud to a stranger, which no guide can do, and that is its distinctive contribution to the first silence. It is not the only component that touches it — the child track is a public-vocabulary intervention entered without stigma (§4.8), and the guides prohibit the euphemisms that keep the vocabulary empty. What the forum adds is witness.

Table 3 distinguishes what the prototype demonstrates from what is designed but unbuilt — a distinction we think matters more than a feature list.

4.4 The role of AI

The assistant's success condition is *a completed handoff to a person* — not a resolved question, and specifically not a returning user. A response that is accurate, warm, and terminates in itself is a failed response.

Three modes are specified. **Ask** locates and surfaces a passage of a guide, and is specified to retrieve only over guides that have been signed off. **Find** translates ordinary language ("someone who can come

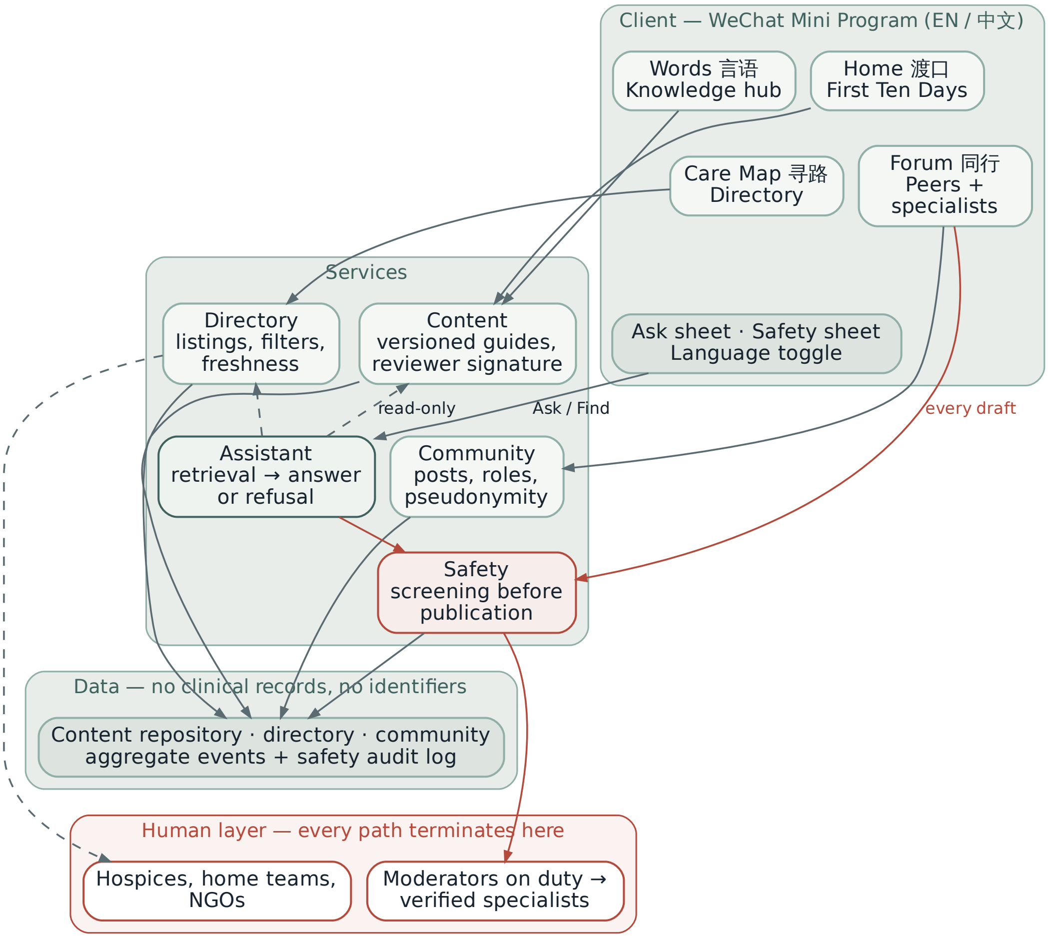

Figure 3: Functional architecture. Four client surfaces and two persistent utilities sit above a service layer, a deliberately thin data layer, and a human layer in which every path is designed to terminate. The assistant has read-only access to guides and listings and no access to community data.

to the house, covered by insurance") into a filtered view of the care map. **Rehearse**, designed but not built, would return scripted openings and let the user practise against fixed text; it is the feature with the clearest path to becoming the substitute relationship §4.1 forbids, since a generative partner playing the dying parent is a role-play with someone about to be bereaved. Our position is that it ships non-generative or not at all.

Four technical commitments follow (Figure 5). The assistant retrieves over a *closed corpus* of signed-off guides and verified listings only. It *refuses by default* when retrieval does not support an answer, saying so plainly rather than improvising. It *restates no clinical content*, surfacing the source instead, because here provenance is part of the meaning. And it has *no persona, no memory, and no ability to initiate contact*: no name, no simulated feelings, no streaks, no notifications. That is a deliberate forfeit of every mechanism that makes companion applications retentive, and it follows from the model in §7 — the desired trajectory is a user who reads two guides, makes a call, has a conversation, and does not come back.

In the current prototype the assistant is a deterministic keyword matcher over pre-written answers with a labelled hook for a live model, chosen so that a demonstration to a clinician or a grieving family cannot fabricate clinical content. It is separately evaluable before any user study, on groundedness,

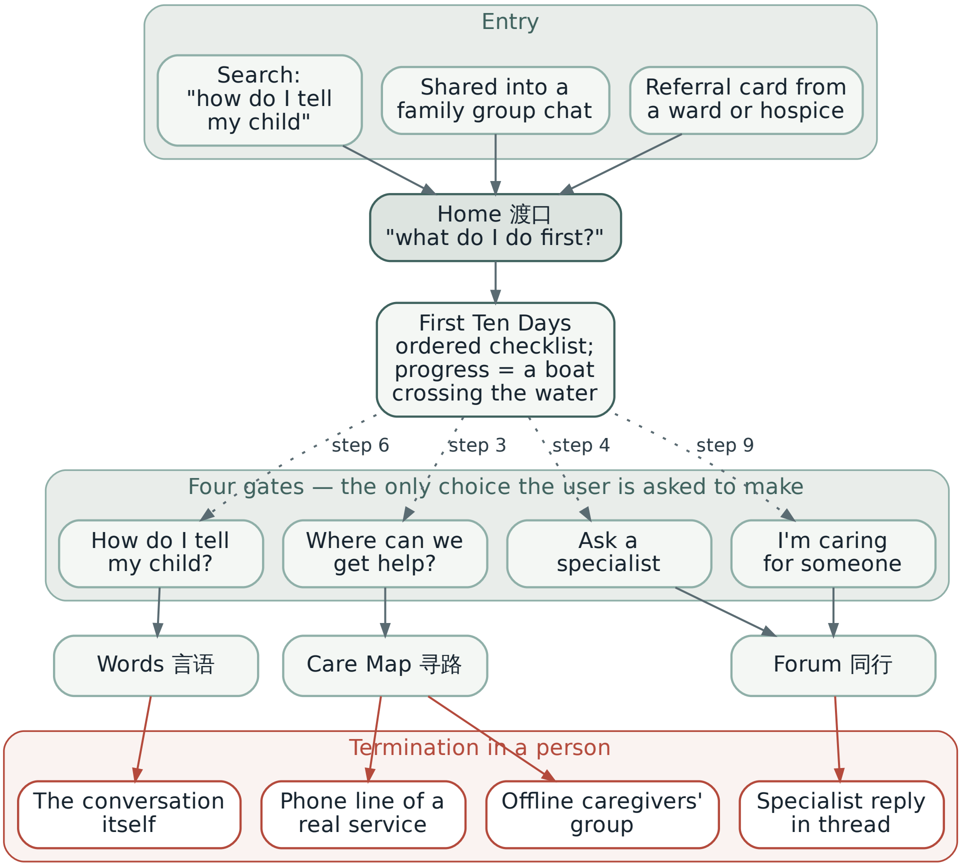

Figure 4: Navigation model. The checklist carries the sequencing, so the user is never required to know the name of what they need; the four gates are the only branching decision asked of them.

Table 3: Implementation status. The prototype is a design and content artifact, not a deployed service; no component has been clinically signed off.

Component	Status in the prototype
Ten-step checklist, gates, three-silences framing	Implemented, interactive
Age-graded child guides; adult conversation guide	Implemented as content
Care map with filters and listing detail	Implemented; labelled sample data
Forum sections, badges, pseudonymity, threads	Implemented; sample posts, no backend
Assistant: Ask and Find	Deterministic sample engine; documented hook for a live model
Assistant: Rehearse	Designed, not built
Specialist verification; secondary hub articles	Designed; content unwritten
Risk screening and crisis routing	Protocol and interface represented; detection logic deliberately absent
Bilingual EN / 中文 parity	Implemented on every screen
Ritual and seasonal surfaces	Proposed only (§4.5)

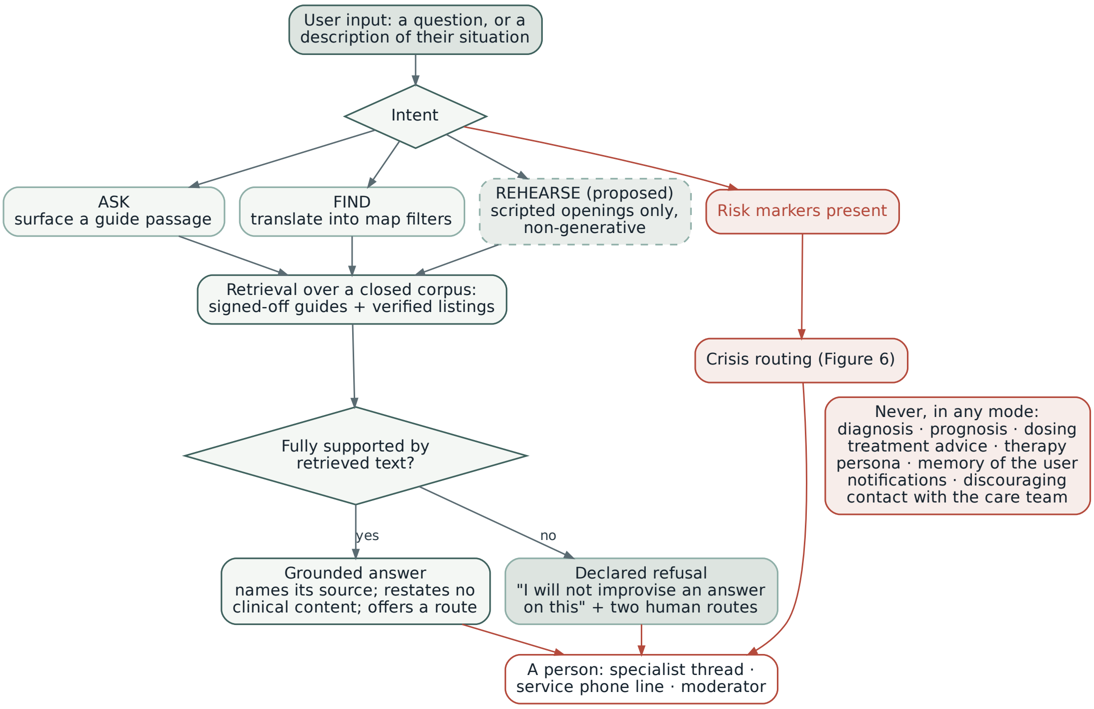

Figure 5: Assistant request pipeline, with the prohibition set shown alongside the grounding gate. The dashed mode is designed but not implemented.

appropriate-refusal rate against an adversarial set, routing completion, and repeat-query consistency.

4.5 Cultural adaptation and affective design

Euphemism versus honesty. The platform holds two positions at once: *the interface is gentle; the content is honest.* Surrounding copy uses the crossing metaphor and ordinary language; the guides inside prohibit the euphemisms that cause harm. No screen tells a reader that their mother is dying. The child guide tells them to say exactly that to a six-year-old.

Naming rather than maintaining the pretence. The platform neither sustains mutual pretence nor attacks it. It names it, explains why speaking first feels like betrayal under filial obligation, and argues that naming is usually kinder than sustaining. This is the difference between instructing and giving permission; where avoidance is culturally structured (Tu et al., 2022; Cheng et al., 2019), permission is the more plausible mechanism.

Terminology. Because the field's own vocabulary varies by region and institution, search and the care map index all variants, so the user never has to guess which term their city uses — a small feature aimed directly at a documented awareness gap (Tang et al., 2025).

The crossing as progress. Checklist progress is drawn as a boat moving across water, leaving a wake. The alternatives encode the wrong thing: a streak says *come back tomorrow*; a percentage says *you are 40% finished*, which is grotesque applied to a dying parent. A crossing says: this has two shores, you are somewhere in it, and it ends.

Ritual codes (proposed). A seasonal surface at Qingming, when grief is socially licensed in a way it is not for the rest of the year — surfaced only if the user opens the platform, never pushed, since a notification that arrives on the anniversary of a death is precisely the re-engagement mechanic Table 2 forbids; a lantern as the saved artifact of 四道人生 and the legacy modules, shareable back into the family group chat; and a low-luminance night mode justified by the caregiver's actual reading hours —

between one and five in the morning, at a bedside, next to someone sleeping.

Access. In the most recent national survey of older adults' living conditions only about a third reported knowing how to use a smartphone (Yicai Global, 2024), which is both why the primary user is the adult child and why the interface must survive being handed to a parent. Large type, generous tap targets, no timed interactions, no video dependence, offline reading, and no account required for the checklist or guides. Chinese and English are to be authored in parallel, with Chinese primary; in the current prototype the Chinese was written from the English and awaits review by a native reader.

4.6 Safety, ethics, and data governance

A forum about dying will contain suicidal statements, requests for medical advice, and grief that other users cannot hold. Safety is not risk mitigation appended to the product; it is the condition under which the forum may exist. If the safety design fails, the correct decision is not to ship the forum.

Four properties define the crisis protocol (Figure 6). Screening happens *while text is being written*, before publication rather than after a report. Risk is *tiered*, with no intervention for distress without markers, because pathologising ordinary grief teaches users that honesty triggers machinery. The post is *not blocked*: suppressing an honest post teaches the author that the one place they were honest punished them for it, so the support card is additive rather than substitutive. And every firing is *escalated and audited*, with automatic escalation if no moderator responds inside a published window.

The marker list, the response windows, and the helpline numbers are deliberately *not specified here* and are labelled placeholders in the prototype. They must be written and signed by clinical members of the team against the National Health Commission's hospice care practice guideline, whose 2025 edition took effect on 26 August 2025 and repealed the 2017 trial version (National Health Commission of the People's Republic of China, General Office, 2025), and against current mental-health referral routes, and validated by expert consensus (§8.2, Phase 0). Publishing an unverified crisis number would be a worse failure than publishing none.

We should be exact about what Phase 0 delivers, because it is easy to overstate. An expert review and a two-round Delphi produce a clinically agreed *marker set*. They do not produce a validated classifier with known sensitivity and specificity over Chinese-language free text, which is a separate research problem with its own annotation scheme, inter-rater work and held-out validation set. Detector development is therefore a distinct workstream with its own gate, not a downstream detail of Phase 0, and until it is complete RQ4's operating characteristics are answerable only for human-flagged material and for the routing interface itself.

Moderation implies a staffing model that is the platform's real scaling limit. Specialist verification requires licence checks and a scope statement; moderators require training, a rota, and supervision, because moderating this material is itself distressing, and a platform that burns out its moderators has reproduced the burnout it exists to reduce.

Naming that risk is not the same as managing it, so we commit to the operational form. Moderators work a capped shift on the crisis queue with mandatory rotation off it afterwards; clinical supervision is scheduled rather than available on request; a named duty clinical contact holds escalation and is not the same person as the moderator on shift; and the forum's growth rate is bounded by the rota rather than by demand. We hold ourselves to the standard we apply to users: if the staffing model cannot be met, the correct decision is not to ship the forum.

One boundary follows from all of this: minors are not users. This is a stated scope and a terms-of-use position rather than a technical control — a mini program shared into a family group chat cannot verify age, and we do not claim otherwise. The consequence is that the child-facing material is written for an adult to read and use, never addressed to a child directly, and that the forum's sections are adult-facing throughout. That is also what allows us to publish detailed guidance about children without operating a service for them.

Two consequences we would rather state than discover. Legally, Article 31 of the Personal Information Protection Law treats the personal information of minors under fourteen as sensitive and requires guardian consent (Standing Committee of the National People's Congress, 2021); a service that cannot verify age cannot rely on a terms-of-use exclusion alone, and the mitigation is architectural — the forum requires no personal information to read, and the child-facing material collects none at all. Practically, the foreseeable failure is specific rather than general: the grief sections will attract bereaved adolescents, and the child guides will be searched by teenagers who have lost a parent. Silence is the wrong response from a platform built against silence, so the sections carry a signposted redirect to youth bereavement services and school counselling routes rather than nothing.

Data governance follows a single principle (Figure 7): the platform should be able to help without learning who the user is or who is dying. Progress and saved cards stay on the device; posts are pseudonymous; assistant queries are transient and unlinked; analytics are aggregate with no free text. We collect no clinical records, real names, addresses, contacts, precise location, or photographs, and we neither advertise nor make user content available for third-party model training. The safety audit log is the most sensitive data we hold and is scoped accordingly.

The scope of that guarantee should be stated precisely, because it is narrower than the sentence above sounds. It holds for what the platform collects and retains. It does not extend to the channel: a WeChat Mini Program necessarily transmits to the operator, whose review rules and data handling we do not control and have already named as a dependency in §4.3. Pseudonymity on our side is not pseudonymity in the stack, and any deployment would require an operator-level data protection impact assessment before, not after, launch.

4.7 Scope: what we cover, and what we do not

An intervention aimed at a taboo is unusually prone to overclaiming, and overclaiming here has a specific danger: a platform that makes a family feel they have addressed the situation, when what they have actually done is read an article, is another way of repairing the story. So we state the boundary explicitly (Figure 8).

In scope: supplying the words; sequencing the first fortnight after a diagnosis; making services findable; providing a place where the words are said out loud; and routing — not counselling — via a constrained assistant.

Roadmap: non-generative Rehearse; open posting at scale and regional sections; seasonal and ritual surfaces; teleconsultation handoff where regulation permits. These are deferred until moderation capacity, clinical sign-off, or evidence exists, and we would rather ship four working surfaces than eight half-supported ones.

Out of scope, and not solvable by an app: clinical care and symptom management; service supply, beds, and workforce training; insurance coverage rules; the disclosure decision itself, which belongs to clinicians and the family together; psychotherapy and treatment of complicated grief; and legal instruments such as advance directives — a boundary that is becoming sharper rather than softer, since Article 78 of the revised 《深圳经济特区医疗条例》, passed in June 2022 and in force from 1 January 2023, made Shenzhen the first jurisdiction in mainland China to give living wills legal effect (Fu et al., 2024), and the platform must therefore route to a legal process rather than describe one. For each of these the platform's only legitimate role is to route to whoever can act. Naming them matters because our own conceptual model (§7) predicts that the navigation components will do nothing at all where no service exists — which is a policy problem wearing the costume of a product problem.

4.8 Dissemination, and the limit that governs it

For a product that touches a taboo, consumer advertising is not merely ineffective but counterproductive: an advertisement for help with dying, arriving unbidden, is an intrusion. The platform carries no advertising of any kind (§4.6), so reach has to be earned rather than bought. Four routes, in the order they can be built, and one constraint that governs all of them.

Reach. Entry is through the topics people already search for and can admit to searching for: how to talk to a child about death, grief that has outlasted other people's patience, the practical mechanics

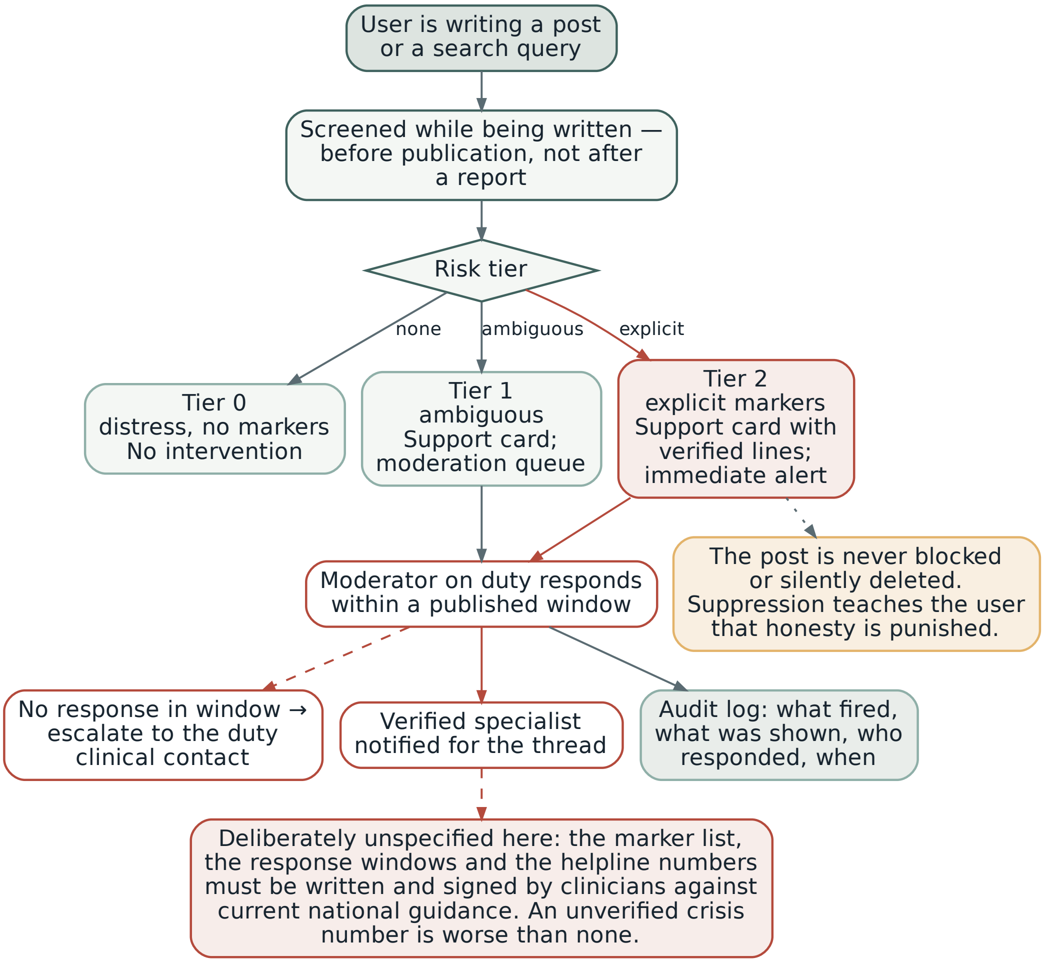

Figure 6: Crisis routing protocol. Screening precedes publication, the post is never suppressed, and the detection parameters are deliberately left to clinical authorship rather than specified by us.

of caring for someone at home. These are the low-stigma doors of §5, and the reason the child track exists. The channels are the ones Chinese readers use for material of this kind — Official Accounts for long-form, short-video and lifestyle platforms for the entry pieces — and the moments are those when remembrance is already socially licensed: Qingming above all, and Zhongyuan. A campaign anchored to those dates says nothing the day has not already said.

Legitimacy. The strongest channel is not a channel: it is a nurse telling a family that something exists. Partnerships with palliative, oncology and nursing departments, with hospices and with NGOs, do double duty, since the same relationships verify the care map (§4.3) and supply the verified specialists (§4.6); university nursing and psychology departments anchor the content. But §8.4 constrains how this may be done. Material placed where a patient may encounter it is a disclosure event that no one chose. Materials therefore pass from clinician to family by hand rather than being displayed in waiting rooms, and anything displayed describes family communication and caregiving without asserting terminality.

Access. Distribution as a Mini Program removes installation entirely and lets the platform arrive the way care is already coordinated — as a link in the family group chat. Printed guides carry codes into workshops and community sessions, subject to the constraint above. Offline gatherings, including the card-based death-education formats already piloted in China — a series of life-and-death education

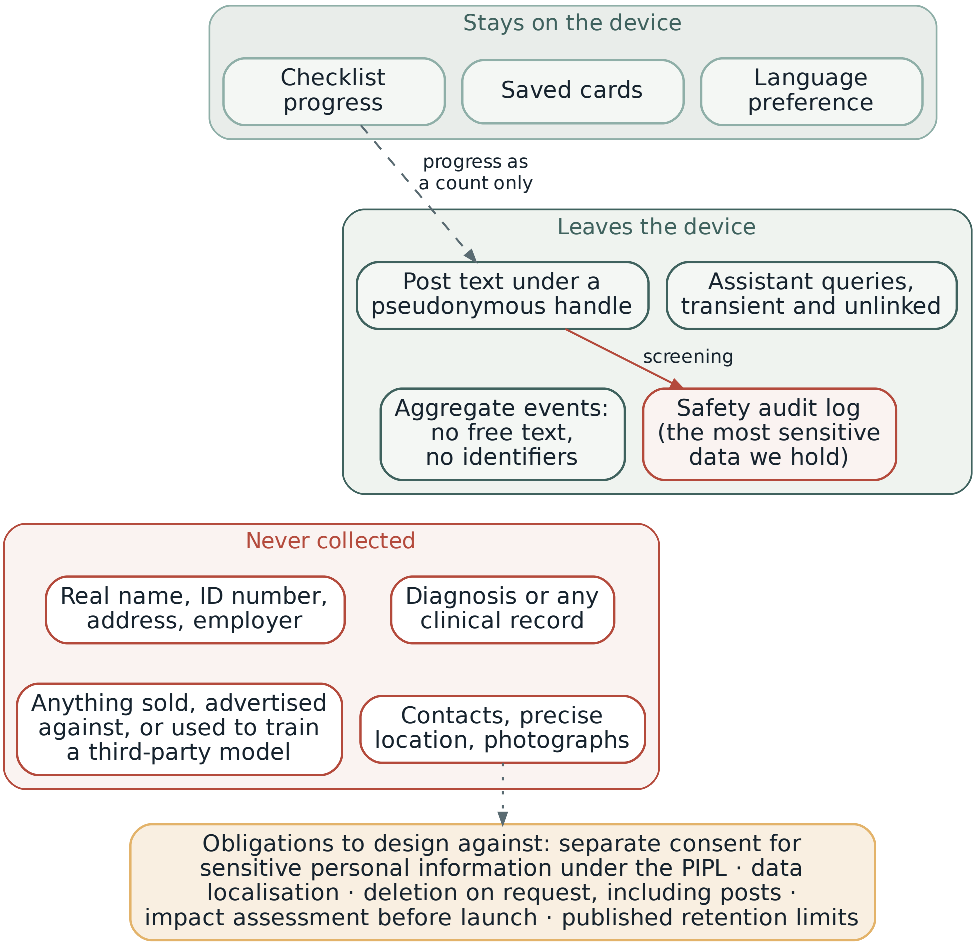

Figure 7: Data minimisation model. The design intent is that the platform can help a user without learning who they are or who is dying.

card games developed from 2021 and run with facilitators in universities and communities (China Daily, 2022) — convert a conversation in a room into a resource that survives it.

Circulation, not retention. The platform is built not to hold people (§4.4), so it cannot grow by returning users, and this is a forfeit rather than an oversight. What it can do is travel. Script cards and saved artifacts are made to be forwarded into a family group chat; caregivers pass it to other caregivers; hospice volunteers act as intermediaries. We propose no notifications, no streaks, no scheduled check-ins, and no success measure that counts a returning user.

Geography, and the governing limit. Rollout begins where provision exists, because §4.7 and the model in §7 both predict that navigation does nothing where there is nothing to navigate to; national reach follows online once the map has depth in one or two cities. Two limits bound the whole strategy. The digital divide means the primary user remains the adult child — mitigated by family-mediated use and by community partnership, not solved. And moderation capacity, not channel reach, is the real ceiling: the forum may grow only as fast as the duty rota that keeps it safe (§4.6). A dissemination plan that outruns its moderators has produced the harm the platform exists to reduce.

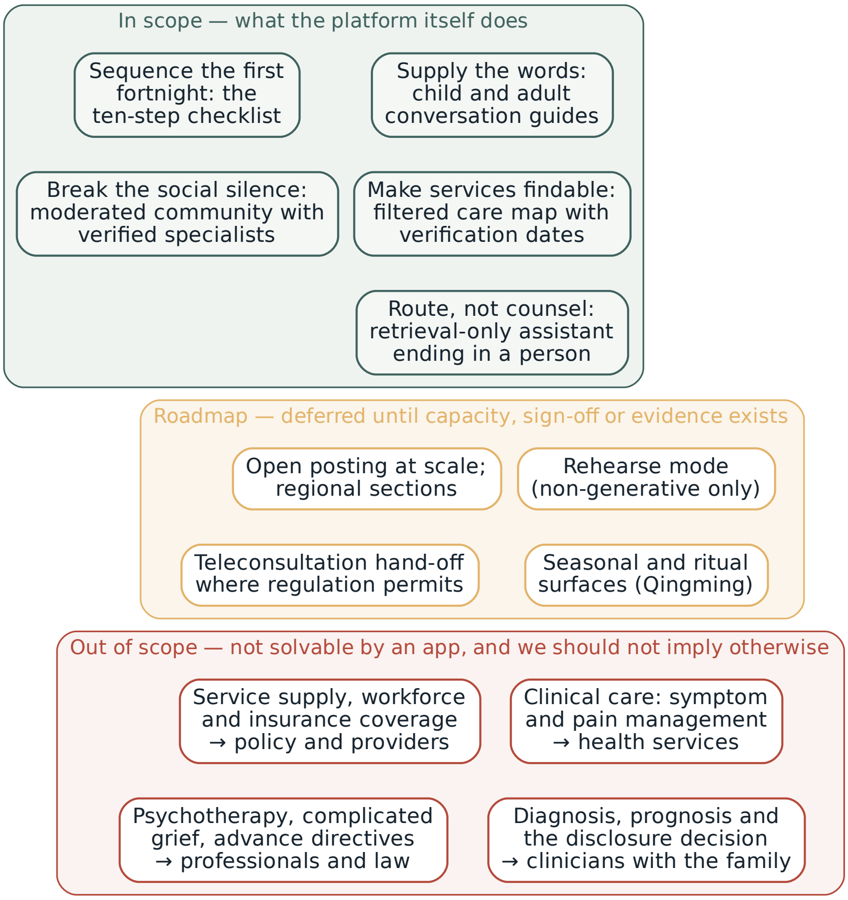

Figure 8: Scope boundary. The dependencies we cannot control run downward: the care map is only as useful as the services that exist, and the assistant refers disclosure decisions out to the people entitled to make them.

5 Personas and Use Scenarios

The people around a dying person are not one audience. Each of them stands in a different relation to the silence, and the platform reaches them through different doors (Table 4). These personas are derived from the literature and from the design process, which means they are hypotheses — and §6 is how we intend to find out how wrong they are.

Two design consequences fall out of the table. First, the primary user is usually *not* the patient, so the platform must be useful to someone acting on behalf of a person who has not consented to the acting — which is an ethical posture, not only a UX one. Second, the lowest-stigma door (a parent explaining a pet's death) is not the highest-need door, and we should expect the population that arrives to be systematically different from the population we most want to help. Both points are tested in §6 and §8.2.

6 Formative Study: Listening to Older Chinese Adults and Their Families

Table 4: Personas, entry points, and the component that serves each. Names are illustrative composites, not participants.

Persona	Where they enter	What they need, and from which surface
The son who was away. Adult child, 38, works in another province; his father has months to live and does not know it.	A link forwarded into the family group chat	Sequencing, above all: what to do first, whom to ask, what to stop deferring. Home checklist → Care Map → the adult conversation guide. He is the primary user and the person holding the secret.
The daughter-in-law who is doing the care.	A search for practical home-care questions	Recognition and relief. She does the work and holds the least authority in the family hierarchy. Peer forum → caregiver sections → support for herself (checklist steps 9–10).
The patient. 71, suspects the truth, asks nobody.	Arrives sideways, via a relative's phone or a shared card	To be asked something real, and to have the option of speaking without forcing a confrontation. The adult guide read from the other side; 四道人生.
The parent of a young child. No one is dying; a grandparent or a pet has died.	Searches "how do I explain death to my child"	Words for one conversation. Age-graded child guides. This is the platform's lowest-stigma door, and the reason the community and map are populated before a family needs them urgently.
The bereaved. Seven months on; people stopped asking at month three.	Returns, or arrives at Qingming	Somewhere the grief is not an imposition. Grief section; bereavement listings. Explicitly not shown as a default to caregivers of living patients.
The village doctor. Township clinic; no palliative training; the only clinician the family will ever ask.	A referral resource to hand to families	Something to give, and peer supervision for the cases that keep him awake. Professionals-only section; printable guides.

6.1 Why we will not design from the literature alone

Everything above is an argument built from published work and a prototype built from that argument. The prototype should therefore be understood as a question, not an answer — and the people best placed to answer it are the ones whose death is being discussed.

There is a specific empirical reason for this, beyond good practice. The literature we rely on partly disagrees with itself. The dominant framing is a death taboo (Tu et al., 2022; Cheng et al., 2019); but older Chinese participants in focus group work discussed death, treatment, and preparation readily (Zhang, 2022), and community-dwelling older adults report substantial demand for death education (Lei et al., 2022). If older people are more willing to talk than their families believe, the design implications invert: the bottleneck is not the older person's reluctance but the adult child's assumption of it, and the intervention should be aimed at least as much at giving permission to the family as at preparing the patient.

We also have concrete design questions no paper will answer. Which words do older adults themselves use, and which do they find intolerable? Would they want to be told? Who do they think should decide? Can they read our type at arm's length, and who is actually holding the phone?

6.2 Participants and sampling

We propose a purposive, quota-controlled sample of roughly 45–60 participants across four streams (Figure 9). *Stream A* is the one this section exists for: 20–24 adults aged 60 and over who are *not* currently dying and not recently bereaved, recruited through urban community centres, senior universities,

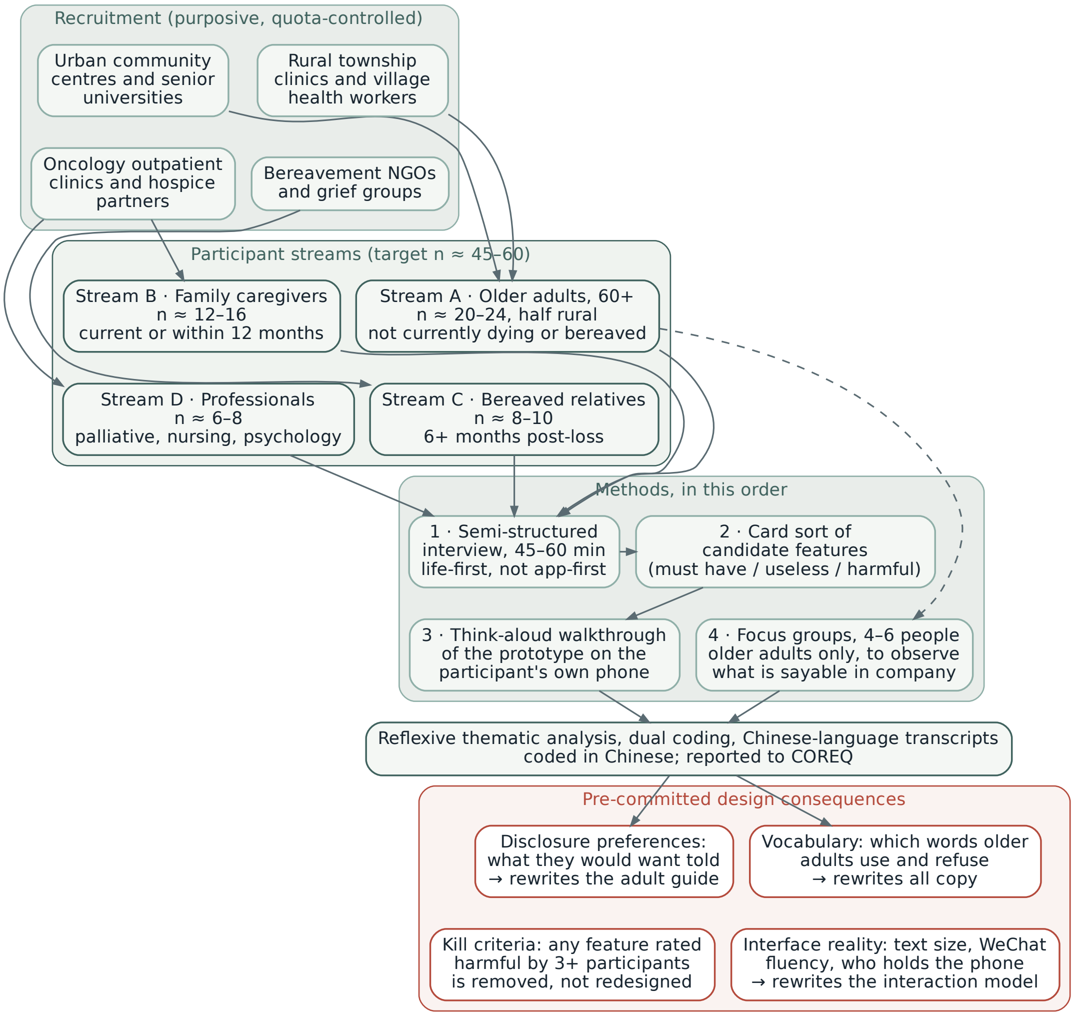

Figure 9: Formative study design. Four participant streams, four methods in fixed order, and design consequences committed to before data collection so that the study can genuinely change the product.

and rural township clinics, with at least half from rural or small-town settings. Sampling older adults who are not in crisis is deliberate: it is the only way to ask what someone would want without asking them to contemplate their own imminent death, and it is also the population whose views their children are guessing at. *Stream B* is 12–16 current or recent family caregivers; *Stream C* is 8–10 bereaved relatives at least six months post-loss; *Stream D* is 6–8 professionals.

Quotas will be set on age band, rural/urban location, and prior experience of a family death, and we will report recruitment refusals, since who declines to discuss death is itself a finding.

6.3 Methods

Four methods in fixed order, per participant where applicable.

(1) Semi-structured interview, 45–60 minutes, conducted in the participant's own language or dialect where possible, and deliberately *life-first rather than app-first*: we ask about experience of deaths in the family, what was said and by whom, what they wish had happened, before we ever mention software. *(2) Card sort* of candidate features, in which participants place cards into must-have, useless, and *harmful* piles — the third pile being the one we most want. *(3) Think-aloud walkthrough* of the prototype on the

participant's own phone, at their own text size, with their own connection, so that we observe the real interaction rather than a laboratory version of it. *(4) Focus groups* of 4–6 older adults only, because what a person will say about death in company differs from what they will say alone, and that difference is precisely our subject matter.

Interviews will be audio-recorded, transcribed in Chinese, and coded in Chinese by two coders to avoid the loss that comes from analysing translations. Analysis will follow reflexive thematic analysis (Braun and Clarke, 2006), with reporting to COREQ (Tong et al., 2007). Feature-level findings will be triangulated against the card sort and the walkthrough behaviour rather than taken from stated preference alone.

6.4 Topic guide domains

Five domains, each tied to a design decision: *vocabulary* — the words used, avoided, and resented, and whether euphemism reads as respect or evasion; *disclosure preference* — whether they would want to be told, by whom, how much, and what should happen if the family disagrees; *family roles* — who decides, who is protected, who is left out; *navigation* — what they know about local services and what they have actually tried; and *technology reality* — what they use WeChat for, text size, who sets up their phone, and what they would never open in front of family.

6.5 Pre-committed design consequences

A formative study that cannot change the product is decoration. We therefore commit in advance to four consequences (Figure 9, right). Vocabulary findings rewrite all user-facing copy, including the terms used in guides. Disclosure findings rewrite the adult conversation guide, and if Stream A participants predominantly say they would want to be told, that finding becomes a prominent element of the guide addressed to families. Interface findings rewrite the interaction model rather than being logged as future work. And any feature that three or more participants place in the *harmful* pile is removed rather than redesigned, unless a clinical reviewer argues otherwise in writing.

6.6 Ethical considerations specific to this study

Interviewing older adults about death carries a foreseeable risk of distress, and a foreseeable risk of something subtler: implying to a healthy person that we have information about their prognosis. Recruitment materials will therefore describe the study as concerned with family communication and views about end-of-life care, never as concerned with the participant's own death. Consent will be obtained in person, in the participant's language, with a plain-language information sheet read aloud where literacy is limited, and with explicit permission sought again before the prototype walkthrough.

Participants may skip any question and stop at any point; a named clinician will be available; and interviewers will be trained to recognise distress and to prioritise it over completing the guide. Where a participant discloses untreated symptoms, unmet care needs, or suicidal ideation, the interviewer follows a written protocol rather than improvising, and the limits of confidentiality are stated at consent rather than invoked afterwards. Interviews will not be conducted in the presence of family members for Stream A, since the presence of the adult child is itself a constraint on what can be said — a methodological point that follows directly from our own theory.

7 Conceptual Model and Mechanisms of Change

Figure 10 states the hypothesised model once. Components act on five mediators — death anxiety and avoidance, communication self-efficacy, perceived norms, knowledge of provision, and perceived isolation — which act on behaviours, which act on distal outcomes, with moderators conditioning the whole chain. Every arrow is a hypothesis.

The assumptions are worth stating plainly, because they are where the model would fail. We assume that reduced avoidance plus increased self-efficacy yields an actual conversation — the intention–

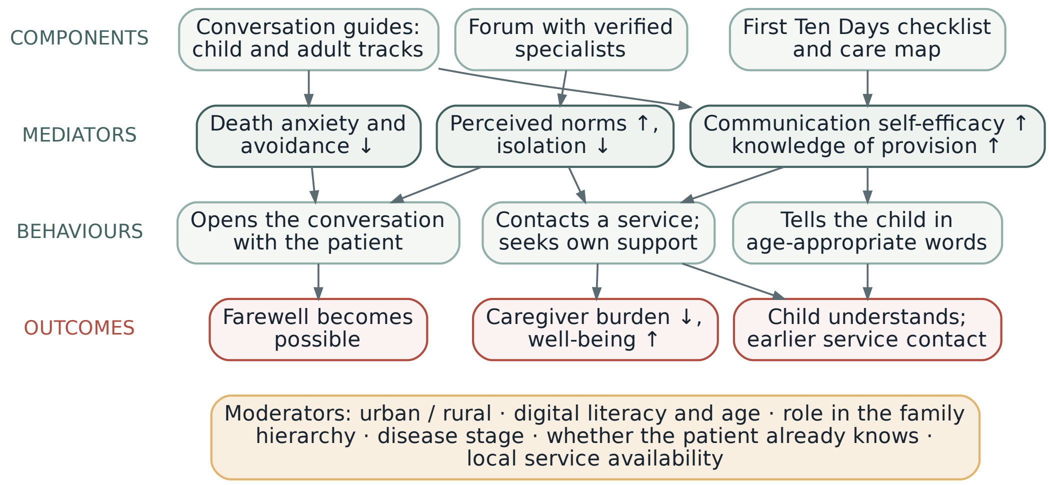

Figure 10: Hypothesised mechanisms of change, simplified into four bands; the grouped mediators are measured separately (§8.3). Mediators draw on terror management (Greenberg et al., 1986), self-efficacy (Bandura, 1997), and the norms component of planned behaviour (Ajzen, 1991); the burden pathway follows stress-process models of caregiving (Pearlin et al., 1990). Nothing here has been tested.

behaviour gap, and the weakest link in the chain (Ajzen, 1991). We assume the conversation helps, though the chain runs through a family exchange we never observe. We assume family-level benefit from a single enrolled user. We assume navigation converts to contact, which requires that something exists locally to contact. And we assume the minimum useful dose is small — one guide read at the right moment — which is what licenses the absence of retention mechanics. That last assumption has a cost worth predicting rather than discovering: digital health interventions lose participants steadily even when they are actively designed to hold them (Eysenbach, 2005), and we have removed every mechanism that normally does the holding. We should therefore expect an attrition profile at the unfavourable end of what that literature reports, and Phase 2b thresholds should be set against that expectation rather than against published rates for retention-optimised products.

The model also makes predictions that can fail, and we prefer to name them now:

- If anxiety falls and willingness rises but reported conversations do not increase, the mediating chain is intact and the behavioural link is broken: the platform needs rehearsal and prompting, not more content.
- If the child track drives adoption with no spillover into caregiving or navigation behaviour, it is an acquisition mechanism and not a therapeutic one.
- If caregiver burden *rises*, the plausible reading is that we successfully transferred responsibility onto an already overloaded person. That is a real harm, and it is a pre-specified safety endpoint rather than a disappointing result.
- If usage concentrates entirely in cities that already have provision, the platform is amplifying an inequity rather than reducing one.
- If users report the platform as a substitute for the conversation — "we read it together, so we didn't need to talk" — then we have built a faster way to repair the story, and the design premise is wrong.
- If death avoidance *rises* rather than falls among participants scoring highest on filial orientation, then mortality salience has strengthened the worldview instead of loosening it, exactly as terror

management research would predict (Burke et al., 2010), and the reframing in §3.2 has failed at the point where it was most needed.

8 Evaluation Framework

8.1 Research questions

RQ1 (attitudes and willingness). Does use of the platform reduce death anxiety and avoidance and increase willingness to engage in end-of-life conversations among adult family caregivers, and by how much? We would also like to know which mediator moves *first*, and are careful about how far the design can answer it: temporal ordering among mediators needs at least three measurement occasions, and preferably more. The schedule therefore carries intermediate waves at two and four weeks on a shortened battery (§8.3), which makes ordering estimable but not securely so. Sequence is accordingly an exploratory question at Phase 3 and a confirmatory one only at Phase 4.

RQ2 (burden and help-seeking). Does the platform affect caregiver burden and help-seeking? Does burden fall, rise, or split by subscale — since better navigation may reduce role strain while more honest conversation initially increases personal strain?

RQ3 (cultural adaptation). Do scripts that name mutual pretence and filial obligation outperform matched direct translations on perceived appropriateness and intention to use, and which adaptation carries the effect? The decomposition is the harder half. A two-arm comparison establishes whether adaptation works in aggregate and cannot attribute the effect to any of its components, so Phase 2a is factorial rather than two-arm (§8.2). We narrow the candidates to the two we believe carry most of the effect — terminology and family framing — and treat ritual codes and visual register as exploratory, because a full factorial over four components is not affordable at this sample size.

RQ4 (safety and the assistant). Does the assistant behave as specified under adversarial conditions, does its presence increase or decrease contact with human support, and at what false-positive rate does the crisis protocol route the cases it is built for? This question becomes answerable only once Phase 0 delivers a signed marker set; until then the assistant's behaviour is evaluable but the crisis protocol's operating characteristics are not.

8.2 Study designs

We have no partner sites, no ethics approval, and no funding at the time of writing, so sample sizes below are illustrative. The phased shape follows established models for developing behavioural interventions, in which efficacy testing comes last (Czajkowski et al., 2015; Skivington et al., 2021), and each phase can stop the programme (Figure 11).

Phase 0 — content and safety validation, no participants: expert review of every guide by a palliative physician, a clinical psychologist, and a paediatric specialist; a two-round Delphi on the crisis protocol; a reading-level check. *Phase 1 — usability and acceptability* ($n \approx 15$–25): think-aloud sessions and interviews following a person-based approach (Yardley et al., 2015), over-sampling older and rural users deliberately, since they are the people this design is most likely to fail and least likely to volunteer. This phase runs jointly with the formative study in §6. *Phase 2a — vignette experiment* ($n \approx 300$–500): the cheapest defensible study in the programme and the one that answers RQ3 most directly, because it tests the adaptation rather than the app. It is factorial rather than two-arm: participants are randomised on terminology (adapted versus matched direct translation) crossed with family framing (naming mutual pretence and 孝 versus an individual-decision framing), following the logic of screening designs for multi-component interventions (Collins et al., 2007). A third factor crosses handoff framing against engagement framing, which is what makes the positional claim in §3.5 falsifiable at a cost we can bear. Ritual codes and visual register are held constant and examined separately, since a full factorial over every candidate component is not affordable at this sample size. Power is calculated on the terminology main effect, for which nothing prevents a calculation now; it will be fixed in the pre-registration before recruitment. *Phase 2b — single-arm pre–post pilot* ($n \approx 40$–60, four to six weeks): feasibility, retention,

Table 5: Each research question, the phase that addresses it, and the measure that operationalises it. The last two rows are claims made elsewhere in the document that a reader is entitled to see mapped to a study, since a proposal should not assert more than its design can reach.

Question	Phase	Design and measure	Status
RQ1 attitudes	2b, 3	Cluster-randomised trial; DAP-R avoidance subscale at six weeks, confirmatory. Waves at 0, 2, 4 and 6 weeks	Sequence exploratory only
RQ2 burden and help-seeking	2b, 3	ZBI-12 by subscale; GHSQ and reported service contact	Secondary
RQ3 adaptation	2a	Factorial vignette: terminology × family framing; appropriateness and intention to use	Confirmatory within 2a
RQ4 assistant and safety	0, 1, 2b	Groundedness coding, adversarial refusal set, audit-log analysis	Crisis operating characteristics deferred to detector workstream
Bridge position, §3.5	2a	Third factor: handoff framing versus engagement framing; intention to contact a human service	Confirmatory within 2a
Conversation occurred	3	Hierarchically ordered secondary, with specificity probes and a dyadic subset	Not confirmatory; §8.2 gives the reasons

adherence, adverse events, and variance estimates (Bowen et al., 2009), with thresholds set in advance that determine whether a trial happens at all.

Phase 3 — controlled trial, conditional on Phase 2b. The obvious design is an individually randomised waitlist trial, and we think it is the wrong one: the intervention is a mini program shared by a link into a family group chat, so contamination between arms is one tap, and caregivers of dying relatives talk to one another. We commit to cluster randomisation by recruitment site rather than leaving the choice open. A stepped-wedge rollout has the practical attraction that every site eventually receives the intervention, but it requires modelling a secular trend and assumes no time-by-intervention interaction — a strong assumption in a policy environment that has changed with each pilot round since 2017. Parallel cluster randomisation buys a cleaner contrast at the cost of that attraction, and we would rather pay it. Blinding participants is impossible, so outcome assessment should be independent, and reporting follows CONSORT-EHEALTH (Eysenbach, 2011). The primary endpoint is death avoidance at six weeks, measured on the avoidance subscale of the DAP-R, chosen because it has a validated Chinese form and because §3.1 argues that reducing avoidance without reducing fear would still count as the intervention working. Communication self-efficacy is theoretically the better endpoint, since §7 predicts it moves first, and it becomes the primary endpoint if and only if the adaptation and validation study flagged in §8.3 is completed before Phase 3; until then it is a named secondary.

We should confront the tension a reader will notice, because it is real. This document repeatedly asserts that the question that matters is whether the conversation happened, and then nominates an *attitude* as the confirmatory endpoint. We have chosen to keep the attitudinal endpoint confirmatory and to justify the choice rather than to disguise it. Three reasons. Death avoidance has a validated Chinese form and known variance, and reported conversation does not. The behavioural measure is self-reported by the person instructed to have the conversation, so demand characteristics are severe. And the advance care planning literature suggests that mediators move on a scale of weeks where behaviour may not (Morrison et al., 2021), meaning that a six-week behavioural endpoint risks being under-powered by construction rather than by chance.

Reported conversation is therefore a *hierarchically ordered secondary*, tested only if the primary is met, and it is protected against the obvious inflation: rather than a yes-or-no item we ask when it happened, where, who was present, and what the opening sentence was, since fabrication survives a single question and rarely survives four. A dyadic report is collected in the subset where the patient is aware of the prognosis and consents, and outcome interviews in the qualitative strand are conducted blind to allocation. If the group later elects to make conversation a co-primary, the multiplicity correction follows and the sample size rises accordingly; we state the trade rather than take it silently.

Every other mediator and every distal outcome is exploratory, and a target effect size will be set from the Phase 2b variance estimates rather than assumed now. *Phase 4* adds bereavement follow-up at 6–12 months and an implementation study of whether the care map changed referral patterns.

Death during follow-up. The endpoint sits at six weeks in a cohort of caregivers of terminally ill relatives, so a meaningful proportion of index patients will die inside the window, and the protocol has to say what happens then rather than discover it. Three things change at once: the caregiver moves from anticipatory to acute grief, which alters the construct the DAP-R is measuring for that person; the Zarit items become partly inapplicable because the caregiving role has ended; and any difference in death rates between arms would bias the comparison. We therefore pre-specify that the primary analysis is conducted on all randomised participants with the six-week measure attempted regardless of bereavement status, accompanied by a pre-registered sensitivity analysis restricted to those not bereaved during follow-up, with bereavement in the window recorded as a covariate and reported by arm. Where the two analyses disagree, the sensitivity analysis is the one we would believe, and we say so in advance rather than after seeing which is more favourable.

Analysis population and missing data. The primary analysis is by intention to treat, with allocation defined at the level of the recruitment site and no exclusion for non-use. Exposure is genuinely hard to observe here, because the platform requires no account and holds progress on the device: we therefore treat allocation as the exposure of record and use self-reported use only as a descriptive covariate, rather than constructing a per-protocol population from a measure we cannot verify. Missing outcome data are handled by multiple imputation under a missing-at-random assumption, with a tipping-point analysis reported alongside, since attrition in a bereaved and exhausted population is unlikely to be ignorable. Mediation is estimated within a counterfactual framework with mediators measured before the outcome wave, and is reported as exploratory throughout.

Contamination is also the delivery mechanism. Our reason for rejecting individual randomisation is that a link shared into a family group chat crosses arms in one tap. That reason cuts both ways, and we should say so: the dissemination model in §4.8 is built on precisely that tap, on circulation rather than retention. A design that successfully suppresses spread is therefore estimating the effect of a version of the platform we do not intend to deploy, and will probably under-estimate what happens in the field. The trial answers an individual-level question under artificially suppressed diffusion; spread is measured rather than controlled away in the Phase 4 implementation study, and neither on its own is the whole answer.

With one confirmatory endpoint and every other measure exploratory, no formal multiplicity correction is required for the primary comparison; exploratory analyses will be reported as such, with effect estimates and intervals rather than significance claims, and the pre-registration will fix this division before data collection.

Both designs carry a cost we should name: with a realistic number of partner sites, cluster and stepped-wedge designs are frequently under-powered, and the design effect depends on an intracluster correlation we cannot yet estimate. Phase 2b should therefore return not only variance but a defensible ICC assumption, and if the achievable number of clusters cannot support the primary comparison, the honest conclusion is a larger multi-site collaboration rather than an underpowered trial.

A qualitative strand runs through every phase, because the outcomes we care most about — a farewell that became possible, a child who was told — will not appear in a mean difference.

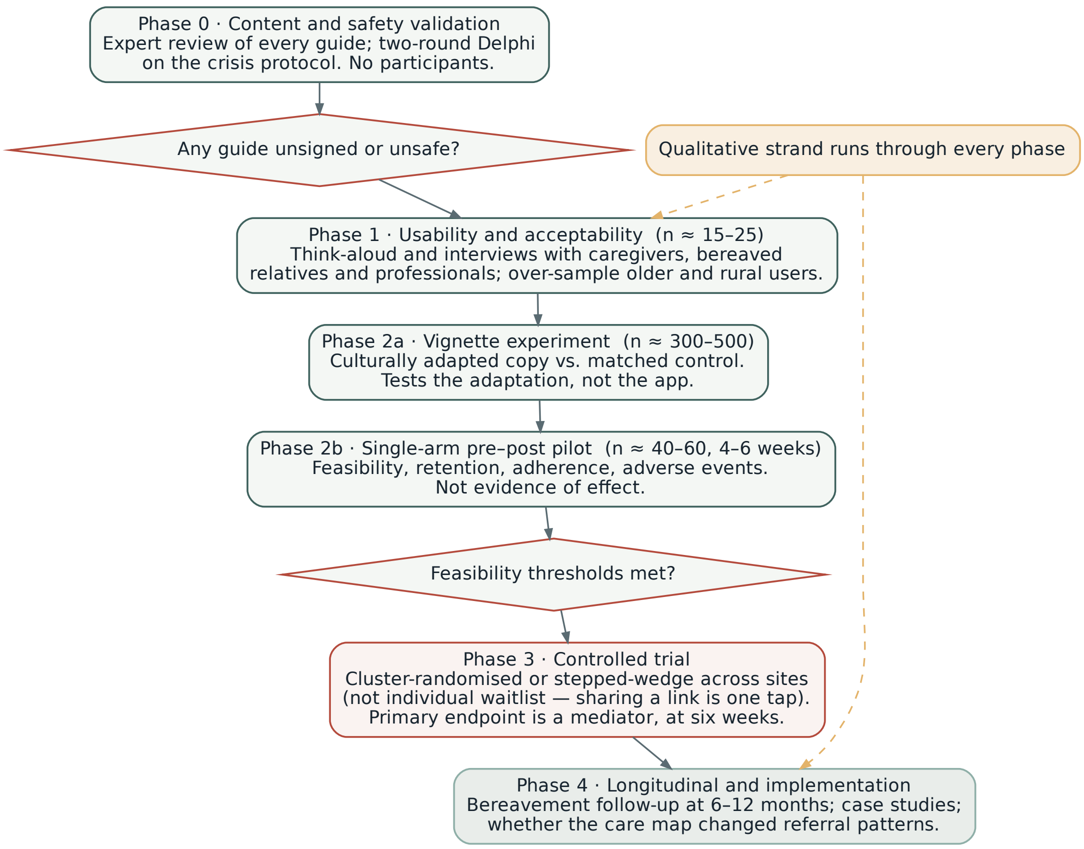

Figure 11: Proposed evaluation sequence, with go/no-go gates. No phase assumes the previous one succeeded.

8.3 Outcome measures

Instruments are chosen for the constructs in §7, favouring short forms with validated Chinese versions (Table 6). Battery length is itself a design constraint: a caregiver in the fourth week of a terminal illness will not complete forty minutes of questionnaires, so we cap the battery at about fifteen minutes.

An earlier draft of this table violated that cap. Administered in full the instruments below run past a hundred items, which is nearer thirty-five minutes than fifteen for a distressed or older respondent, and the constraint would have been broken on the same page that stated it. The battery is therefore specified rather than listed. At the full waves we administer the DAP-R avoidance and fear subscales rather than the complete profile, the knowledge block of the P-KAPHPCs rather than the whole scale, ZBI-12, PHQ-4, the help-seeking items and the purpose-built intention items: 62 items, which pilots at eleven to thirteen minutes. The eHEALS and SUS are collected once, at Phase 1 and at the end of Phase 2b, and never at an outcome wave. The intermediate waves at two and four weeks carry the avoidance subscale, the self-efficacy items and the intention items only: 21 items, under four minutes. Where a wave would exceed the cap, the cap wins and the instrument is dropped, not the other way round.

Two honest gaps. No validated Chinese end-of-life communication self-efficacy measure was identified, so adaptation and a small validation study would be needed. And the three outcomes we most want — the patient's experience, the child's adjustment, the quality of a farewell — have no acceptable direct measurement route in an early-phase study, and would be approached through proxy report and qualitative interview rather than pretended into a scale.

Table 6: Candidate outcome measures. Engagement analytics are never a primary endpoint, for the reason given in §4.4.

Construct	Instrument	Phase
Death attitudes and avoidance	Death Attitude Profile–Revised (Wong et al., 1994); avoidance subscale primary, fear subscale secondary	2b, 3
Knowledge and attitudes to palliative care	Public Knowledge, Attitude and Preference of Hospice and Palliative Care Scale (P-KAPHPCs), developed and culturally localised for China by Tang et al. (2025); already Chinese, so no translation is required, but it was built for a single survey and would need the authors' permission and a psychometric check in our sample	2b, 3
Communication self-efficacy	No validated Chinese measure identified; adaptation required	2b, 3
Willingness to talk	Purpose-built intention items; vignette-based willingness	2a, 2b
Help-seeking intentions and behaviour	General Help Seeking Questionnaire, adapted (Wilson et al., 2005); plus reported service contact	2b, 3
Caregiver burden	Zarit Burden Interview, 12-item version (Zarit et al., 1980; Bédard et al., 2001)	2b, 3
Caregiver distress	PHQ-4 (Kroenke et al., 2009)	2b, 3
Complicated grief	Inventory of Complicated Grief (Prigerson et al., 1995)	4 only
Usability; acceptability	SUS (Brooke, 1996); Theoretical Framework of Acceptability (Sekhon et al., 2017)	1, 2b
eHealth literacy (covariate)	eHEALS (Norman and Skinner, 2006)	1, 2b
Assistant behaviour	Two-rater groundedness coding; adversarial refusal set; audit-log analysis	0, 1, 2b

8.4 Research ethics

Any study requires institutional ethics approval, and site approval where recruitment runs through hospitals or hospices.

The consent problem here is specific, and created by the phenomenon under study. *If a patient does not know their prognosis, a study invitation can break the family's concealment.* A poster reading "for family caregivers of patients with terminal illness," seen by a patient in an oncology waiting room, is a disclosure event that neither the family nor the research team chose. We would therefore recruit caregivers directly and never through the patient, use materials describing family communication without asserting terminality, hold consent conversations away from the bedside, and name inadvertent disclosure as a protocol risk with a documented response. This also sharply limits patient-reported outcomes, which is a real cost rather than an oversight.

Vulnerability follows from the population: dying patients are not enrolled in early phases, children are not enrolled before Phase 4, and bereaved participants are not approached before six months post-loss. Measurement itself can distress, so every study needs a right to skip, a stated support pathway, and an available clinician.

Most importantly, this intervention can cause harm, and the protocol should say what harm looks like. We propose pre-specifying and collecting as adverse events: family conflict after an attempted conversation; disclosure the patient did not want; distress in a child after being told; increased caregiver burden; and any crisis event during participation. A communication intervention that records no adverse events is not a safe intervention but an unmonitored one.

Finally, we are both the designers and the evaluators. We commit in advance to pre-registration of hypotheses and primary endpoints, independent outcome assessment, an external analyst for part of the qualitative work, and publication of null results.

Those four are procedural. The one that actually constrains us is a fifth: an independent data monitoring and safety committee, convened before Phase 2b and standing through Phase 3, with at least one palliative clinician and one methodologist unaffiliated with the design team, holding a written charter and the authority to pause or stop recruitment. For a trial whose pre-specified adverse events include family conflict, unwanted disclosure and distress in a child, the people who decide whether a safety signal warrants stopping should not also be the people who designed the thing generating it. We would rather be stopped by someone else than trust ourselves to stop.

9 Expected Outcomes and Impact

Against RQ1 we expect small reductions in death avoidance and fear of death and small-to-moderate gains in communication self-efficacy and stated willingness, with self-efficacy and norms moving before anxiety, since the platform supplies sentences and social evidence more directly than reassurance. Against RQ2 we expect increased service contact where services exist, increased help-seeking for the caregiver's own distress, and a burden effect that splits by subscale. Against RQ3 we expect adapted material to outperform matched translations, with terminology and family framing carrying more of the effect than ritual or visual elements. Against RQ4 we expect high groundedness at the cost of a measurable over-refusal rate, and regard that trade as correct here.

From the formative study we expect one finding to matter more than the rest: whether older adults' own stated preferences about disclosure diverge from what their children assume. If they do diverge — as Zhang (2022) and Lei et al. (2022) suggest they may — then the most valuable single sentence the platform can carry is not advice but evidence: *people your parent's age, asked directly, mostly say they would want to know.*

The theoretical contribution is twofold: a specified account of how a digital intervention might change a family-level communication norm rather than an individual attitude, a level at which stigma research rarely operates; and an articulated third position for AI in mental health, carrying the falsifiable claim that a handoff-maximising system outperforms an engagement-maximising one. The practical contribution is a transferable framework — the theory-to-feature mapping, the prohibition set, the crisis protocol, and the scope boundary are portable to any setting where a health topic is taboo and services are hard to find. If the platform works even modestly at scale, its public-health value lies less in wellbeing scores than in shortening the interval between a terminal diagnosis and the first contact with palliative care.

10 Limitations

The first limitation is status. This is a prototype and a proposal: no component has been clinically signed off, no study has been run, the care map and forum contain labelled sample content, and every effect above is hypothesised.

The second is that our framing may be too tidy. The two deaths reported in §1 are journalism rather than data: they illustrate a pattern without establishing its prevalence, and a metaphor that organises an argument well can also flatten it. The two cases are also not instances of our subject. Those people died alone because no one was in the room; the families we describe are in the room and silent. What the cases share with protective concealment is the mechanism — shame attaching to circumstances, and repair directed at the account — not the situation. We use them as an image and not as evidence, and a reader who finds the bridge too convenient is entitled to discount the opening without discounting §2 — a caution we now state at the point of use in §1.1 rather than only here. The same applies to two figures that do more work than the cases do: the empty-nest share and the proportion of older adults reporting smartphone competence both reach us through press reporting of the national survey of older adults' living conditions rather than from the survey itself, and the second of them carries a design decision (§4.5).

A third source deserves the same scrutiny. The corrective we lean on in §1.2 and §6.1 — that older people may be more willing to discuss death than their families assume — rests on focus groups with older Chinese *immigrants in Canada* (Zhang, 2022). Migration selects, and a Canadian setting supplies institutional norms of disclosure that mainland participants have not been exposed to, so the transfer to an inland rural county is a real inferential leap rather than a small one. We rely on it heavily enough that it shapes the design, and it is precisely why Stream A of the formative study exists: to replace a borrowed finding with a local one before any of this is built on further. Cultural heterogeneity within China is treated too simply throughout — attitudes, family structure, terminology, and service availability differ substantially between coastal cities and inland rural counties, and a design validated in Chengdu may not transfer. Relatedly, the digital divide means the platform reaches the dying mostly through their children, and not at all in families where nobody is digitally fluent.

Third, the outcomes with the greatest moral weight belong to people we should not enrol, and the consent problem in §8.4 removes much of the patient-reported data an evaluator would want; we are proposing to measure reachable proxies for what we actually care about.

Fourth, residual AI risks persist even under our restrictions: a closed corpus reduces but does not eliminate fabrication, over-refusal has its own cost when a frightened person is turned away at two in the morning, and our claim that this design resists dependency is an argument rather than a finding.

Fifth, and most seriously, the intervention could become the thing it opposes. A family that reads a guide together, feels that the matter has been handled, and never has the conversation has used our platform to repair the account more efficiently than any neighbour could. We have tried to design against this — routing requirements, no retention mechanics, behavioural rather than engagement endpoints — but it remains the most likely way for this project to fail while appearing to succeed, and it is why "did the conversation happen" must stay the primary question.

11 Conclusion and Future Work

We have proposed 渡 (Bridge), a culturally grounded platform aimed at the three silences that leave Chinese families alone at the end of life: a knowledge hub that supplies the words, a care map that makes services visible, a community where the words get said out loud, and an AI assistant constrained to carrying people to other people. We have given the theoretical derivation of each component, an explicit boundary around what software cannot fix, a formative study designed to let older adults revise our assumptions, a model of mechanisms, and stated conditions under which we would conclude we are wrong.

The immediate next steps are not technical. They are clinical sign-off of every guide and of the crisis protocol; the formative study in §6; and partnerships with hospices, oncology services, and NGOs, without which the care map cannot be verified, a trial cannot be recruited, and the promise of routing people to real help cannot be kept.

No application would have reached the woman in Nanjing before the community worker did, and nothing in this proposal would have given her back the afternoon she died in. What a platform can plausibly do is earlier, smaller, and harder to photograph: make it likelier that somebody said the thing out loud while there was still time to say it, so that fewer families are left afterwards with an account to repair.

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